

EduSaaS PDDD - Part 1

Chapters 1 to 5: Foundation and Security Domain

Document Type: Production Database Design Document (PDDD)

Schema: education | Database: PostgreSQL | Architecture: Backend-owned persistence, stateless AI services

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Chapter 1: Executive Summary

The EduSaaS database platform is the backend-owned persistence layer for user identity, learning management, assessments, skill gap analysis, recommendations, community engagement, employer workflows, administration and AI support features. AI services remain stateless and do not own database tables.

Metric	Value
Schema	education
Database	PostgreSQL
Total Tables	46
Existing Backend Tables	41
New AI Support Tables	5
AI DB Ownership	None, backend owns persistence

Chapter 2: Architecture Overview

The platform follows a layered architecture: Frontend -> Backend APIs -> PostgreSQL education schema, with stateless AI services called by backend for recommendations, scoring and prediction activities.

Frontend -> Backend/Spring Boot -> PostgreSQL education schema
Backend -> AI Service/FastAPI -> ML models -> response to Backend

Chapter 3: Database Standards

Tables use plural names and columns use snake_case. Most business entities use UUID primary keys generated with gen_random_uuid(). Skill and quiz tables use SERIAL/INTEGER ids where defined in baseline schema.

Standard	Rule
Table names	Plural lower snake_case
Columns	snake_case
PK	id or explicit entity id
FK	fk_<table>_<reference>
Index	idx_<table>_<column>
Audit	created_at and updated_at where applicable

Chapter 4: Database Inventory

Security & Identity

#	Table
1	roles
2	permissions
3	role_permissions
4	users
5	profiles
6	auth_otps
7	refresh_tokens

Learning

#	Table
1	courses
2	lessons
3	quizzes

4	assignments
5	enrollments
6	progress
7	certificates
8	achievements
9	tasks
10	recommendations

Assessment & Skills

#	Table
1	domain_roles
2	skills
3	domain_required_skills
4	assessments
5	gap_reports
6	difficulty_levels
7	questions
8	quiz_sessions
9	student_answers
10	student_skill_results
11	quiz_state

Employer

#	Table
1	jobs
2	applications

Community

#	Table
1	community_posts
2	community_comments
3	community_reactions
4	community_tags
5	community_post_tags
6	community_bookmarks

Administration

#	Table
1	notifications
2	subscriptions
3	settings
4	reports
5	announcements

AI Support

#	Table
1	activity_logs
2	login_history
3	course_skills
4	course_prerequisites
5	course_ratings

Chapter 5: Security & Identity Domain

This domain manages authentication, authorization, user profiles, RBAC, OTP verification and refresh token lifecycle.

Table: education.roles

Column	Type	Constraint	Description
id	UUID	PK	Unique role identifier
name	VARCHAR(100)	UNIQUE, NOT NULL	Role name
description	TEXT	NULL	Role description
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created timestamp

Table: education.permissions

Column	Type	Constraint	Description
id	UUID	PK	Unique permission identifier
name	VARCHAR(150)	UNIQUE, NOT NULL	Permission name
description	TEXT	NULL	Description
category	VARCHAR(100)	NULL	Permission category
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created timestamp

Table: education.role_permissions

Column	Type	Constraint	Description
role_id	UUID	PK/FK	References roles.id
permission_id	UUID	PK/FK	References permissions.id
assigned_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Assignment timestamp

Table: education.users

Column	Type	Constraint	Description
id	UUID	PK	User identifier
name	VARCHAR(255)	NOT NULL	Full name
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email
role_id	UUID	FK	References roles.id
password_hash	TEXT	NOT NULL	Password hash only
status	VARCHAR(50)	DEFAULT active	Account status
last_login	TIMESTAMP(6)	NULL	Last login
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created timestamp
domain_role_id	UUID	FK NULL	References domain_roles.domain_role_id

Table: education.profiles

Column	Type	Constraint	Description
id	UUID	PK	Profile identifier
user_id	UUID	UNIQUE FK	References users.id
career_goal	TEXT	NULL	Career target
institution	TEXT	NULL	Institution
company	TEXT	NULL	Company
preferences	JSONB	NULL	Preferences
initial_assessment_completed	BOOLEAN	DEFAULT false	Initial assessment flag

Table: education.auth_otps

Column	Type	Constraint	Description
id	UUID	PK	OTP record
user_id	UUID	FK UNIQUE	References users.id
otp_hash	TEXT	NOT NULL	Hashed OTP

expires_at	TIMESTAMP(6)	NOT NULL	Expiry
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
verified_at	TIMESTAMP(6)	NULL	Verified timestamp

Table: education.refresh_tokens

Column	Type	Constraint	Description
id	UUID	PK	Token record
user_id	UUID	FK	References users.id
token_hash	TEXT	NOT NULL	Hashed refresh token
expires_at	TIMESTAMP(6)	NOT NULL	Expiry
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

RBAC Implementation Notes

Roles contain business role definitions; permissions contain API/action permissions; role_permissions maps roles to permissions; users.role_id assigns one primary role to a user. Backend resolves effective permissions after login and includes them in JWT/authorization context.